Kostadin Devedzhiev

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SUMMARY

First-year PhD student at the University of Cambridge's Centre for Human-Inspired AI (CHIA), developing orchestration algorithms for human–AI systems, combining task allocation, skill inference and online learning. My research focuses on pro-human orchestration: optimising how work is distributed while treating human participation as an explicit constraint, not a by-product of maximising output. I developed skillinfer, an open-source Python package, and previously spent three years at Stellar Cyber building human–AI multi-agent security products, demonstrated at RSA and Black Hat.

EDUCATION

University of Cambridge Oct 2026 – Present
PhD in Human-Inspired Artificial Intelligence · Queens' College Cambridge, UK

- Research on how to split work between people and AI agents when they differ in skill, cost and availability.
- CHIA, Trustworthy AI (TRACE) Lab. Co-supervised by Prof. Umang Bhatt and Prof. Adrian Weller.

University of Cambridge Oct 2025 – Jun 2026
MPhil in Human-Inspired Artificial Intelligence · Homerton College Cambridge, UK

- Distinction (82.6/100). CHIA, TRACE Lab. Supervised by Prof. Umang Bhatt.
- Thesis: *Ad Hoc Human–AI Orchestration via Skill Inference*.

Stony Brook University Aug 2018 – May 2022
BS, Computer Science and Applied Mathematics & Statistics Stony Brook, NY

- Specialisation in AI and Data Science · CS Honors Program · GPA 3.89/4.00 · Summa Cum Laude.

University of Hawai'i at Hilo Aug 2019 – May 2020
National Student Exchange Program · Computer Science Hilo, HI

- GPA 3.97/4.00.

PROFESSIONAL EXPERIENCE

Stellar Cyber May 2022 – Sep 2025
Software Engineer San Jose, CA

- Designed and built a human-augmented multi-agent security operations system in which large language model (LLM) agents triage security cases by correlating alerts and generating summaries, and analysts review and act on them. Featured in Stellar Cyber's main demos at RSA 2025 and Black Hat 2025.
- Built the chat interface, session management and chart visualisations for Open XDR Investigator, an LLM copilot that converts natural-language questions into Elasticsearch queries. Demonstrated at RSA and Black Hat 2024.
- Implemented bidirectional WebSocket streaming with parallel data and LLM requests, **reducing average query response time by 70%**.
- Built retrieval-augmented LLM agents for log analysis, product metrics, connector normalisation and data-source classification, with support for chart generation, API calls and script execution.
- Maintained **over 90% test coverage** across owned components. Bug filings **fell by 34%** year on year.

University of Hawai'i at Hilo Mar 2020 – May 2020
Creative Electronic Media Assistant Hilo, HI

- Built an API for a hologram display and developed the Data Visualization Lab website.

Vivansa Jun 2019 – Aug 2019
Software Engineering Intern Sofia, Bulgaria

- Improved CRM front-end components and fixed erroneous database records through root-cause analysis and automated data-cleaning procedures.

RESEARCH EXPERIENCE

University of Cambridge Jul 2026 – Sep 2026
Research Assistant, Centre for Human-Inspired AI (CHIA) Cambridge, UK

- Researched task allocation between people and AI agents through Hybrid Workforces, a collaboration between CHIA and Accenture Singapore.
- Developed Bayesian skill inference to estimate the capabilities of human and AI agents, alongside a Human API that lets agents request input from people through an explicit interface.

Stony Brook University Aug 2021 – May 2022
Research Assistant, Natural Language Processing Stony Brook, NY

- Designed Recursive QA, a tool that converts English requirements into formal specifications by guiding annotators through questions based on constituency parse trees.
- Used affinity clustering based on Levenshtein distance to reduce redundant answer options, achieving **over 80% inter-annotator agreement**. Experienced users completed annotations in about 30 seconds.
- Built the full-stack annotation application, including user accounts, work history and graph visualisations.

University of Hawai'i at Hilo Jun 2020 – Aug 2020
AI Research Assistant Hilo, HI

- Developed a heuristic for estimating IoU/F1 thresholds to control proposal acceptance in real-time CNN-based invasive-species detection from drone imagery.
- Built a human-in-the-loop annotation tool that increased AI assistance as the model improved during training.

PUBLICATIONS

P.-M. Binder, **K. G. Devedzhiev**, A. T. Runyan. *Motional emf generated by squeezing an elliptical conducting loop.* 2020
European Journal of Physics. doi:10.1088/1361-6404/abb066

TALKS & PRESENTATIONS

Talk, Oxford Human–Algorithm Interaction Workshop, Saïd Business School – *Ad Hoc Human–AI Orchestration via Skill Inference*. Jul 2026

Poster, CHIA Annual Conference, Cambridge Union – *Ad Hoc Human–AI Orchestration via Skill Inference*. Jun 2026

PROJECTS & SOFTWARE

Ad Hoc Human–AI Orchestration via Skill Inference – MPhil thesis thesis

- Represented each teammate's skill profile as a Bayesian belief over a skill taxonomy. Because skills are correlated, observations about one skill update beliefs about others, **reducing the number of observations** needed to estimate a profile.
- Released as skillinfer, an open-source Python package on PyPI, with source code on GitHub. Used in Tailor.

Tailor – Human–AI orchestration platform tailorworkflow.com

- Built a prototype visual workflow editor for combining AI agents with human review steps, with four levels of human control ranging from fully automated to human-led workflows.
- Uses *skillinfer* to profile people and agents and route each step to whoever is best placed to do it.

TEACHING & MENTORSHIP

AI Leadership Mentor, UniHawk, Kuwait City – Mentored high-school students in a week-long AI app development bootcamp. Dec 2025

Teaching Assistant, Stony Brook University – AMS 210: Applied Linear Algebra. Weekly office hours and grading. Aug 2020 – May 2021

Linear Algebra Grader, University of Hawai'i at Hilo – MATH 311: Linear Algebra. Mar 2020 – May 2020

Computer Science Grader, University of Hawai'i at Hilo – CS 150: Intro to Computer Science. Oct 2019 – Dec 2019

HONOURS & FELLOWSHIPS

Alignment Fellowship (technical track), Cambridge AI Safety Hub. 2026

MPhil with Distinction (82.6/100), University of Cambridge. 2026

Summa Cum Laude, Stony Brook University. 2022

SKILLS

Research	Human–AI orchestration, multi-agent systems, Bayesian skill modelling, human–AI collaboration, HCI
Programming	Python, TypeScript, JavaScript, C
Machine learning	PyTorch, Hugging Face, scikit-learn, LangChain, LangGraph, pandas, NumPy
Data & infra	PostgreSQL, MongoDB, Elasticsearch, Pinecone, Redis, Docker
Web	React, Angular, Tailwind CSS, FastAPI, Node.js, Express, Flask
Languages	Bulgarian (native), English (fluent)